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A scientist investigated the affinity for oxygen of horse haemoglobin and mouse haemoglobin.

Some of their results are shown in **Table 7**.

Table 7

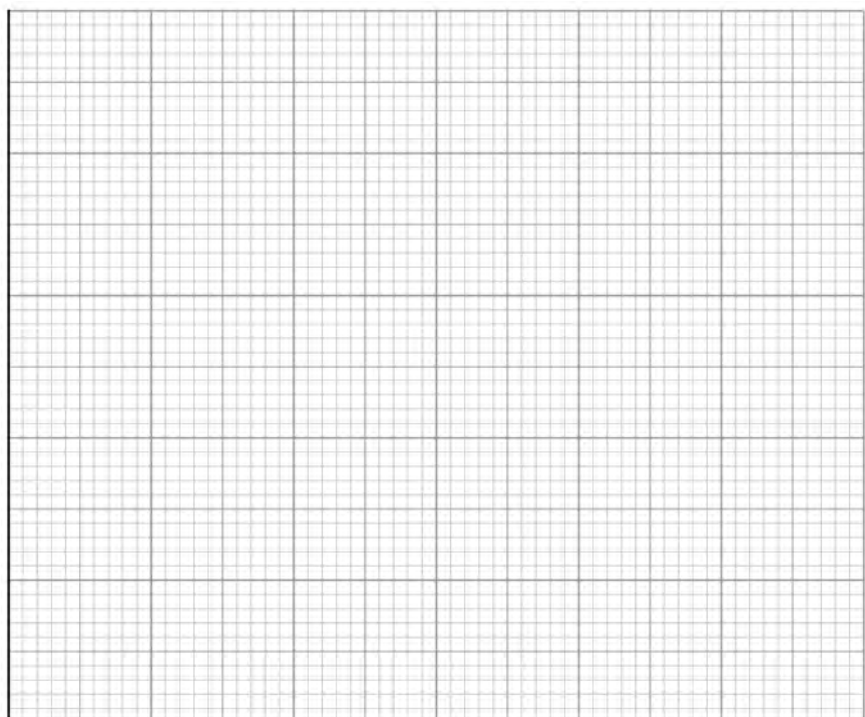
Animal	Partial pressure of oxygen when haemoglobin is 50% saturated / kPa	Partial pressure of oxygen when haemoglobin is 25% saturated / kPa	Body mass of one animal / g
Horse	3.2	1.9	550 000
Mouse	6.5	3.3	23

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Plot the haemoglobin saturation data from **Table 7** and use these points to sketch the full oxyhaemoglobin dissociation curves for a horse and a mouse.

[3 marks]

Percentage
saturation of
oxyhaemoglobin



Partial pressure of oxygen / kPa

0	9	.	2
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The following equation can be used to estimate the metabolic rate of an animal.

$$\text{Metabolic rate} = 63 \times \text{BM}^{-0.27}$$

BM = body mass in grams

Use this equation to calculate how many times faster the metabolic rate of a mouse is than the metabolic rate of a horse.

[2 marks]

Answer = _____ times faster

0	9	.	3
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The data in **Table 7** show differences between the oxyhaemoglobin dissociation curve for a mouse and the oxyhaemoglobin dissociation curve for a horse.

Suggest how these differences allow the mouse to have a higher metabolic rate than the horse.

[2 marks]

[Extra space]

Question 9 continues on the next page

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